Research and Development

We are committed to investing in highly productive research and development capabilities, particularly in electro-mechanical systems. Our research and development ("R&D") expenditures were approximately $48.3 million, $47.3 million and $45.2 million for the years ended December 31, 2017, 2016 and 2015, respectively.

We concentrate on developing technology innovations that will deliver growth through the introduction of new products and solutions, and also on driving continuous improvements in product cost, quality, safety and sustainability.

We manage our R&D team as a global group with an emphasis on a global collaborative approach to identify and develop new technologies and worldwide product platforms. We are organized on a regional basis to leverage expertise in local standards and configurations. In addition to regional engineering centers in each geographic region, we also operate a global engineering center of excellence in Bangalore, India.

Seasonality

Our business experiences seasonality that varies by product line. Because more construction and do-it-yourself projects occur during the second and third calendar quarters of each year in the Northern Hemisphere, our security product sales, typically, are higher in those quarters than in the first and fourth calendar quarters. However, our Interflex business typically experiences higher sales in the fourth calendar quarter due to project timing. Revenue by quarter for the years ended December 31, 2017, 2016 and 2015 are as follows:

	First Quarter	Second Quarter	Third Quarter	Fourth Quarter
2017	23%	26%	25%	26%
2016	22%	26%	26%	26%
2015	22%	25%	26%	27%

Employees

We currently have approximately 10,000 employees.

Environmental Regulation

We have a dedicated environmental program that is designed to reduce the utilization and generation of hazardous materials during the manufacturing process as well as to remediate identified environmental concerns. As to the latter, we are currently engaged in site investigations and remediation activities to address environmental cleanup from past operations at current and former production facilities. The Company regularly evaluates its remediation programs and considers alternative remediation methods that are in addition to, or in replacement of, those currently utilized by the Company based upon enhanced technology and regulatory changes.

We are sometimes a party to environmental lawsuits and claims and have received notices of potential violations of environmental laws and regulations from the U.S. Environmental Protection Agency (the "EPA") and similar state authorities. We have also been identified as a potentially responsible party ("PRP") for cleanup costs associated with off-site waste disposal at federal Superfund and state remediation sites. For all such sites, there are other PRPs and, in most instances, our involvement is minimal.

In estimating our liability, we have assumed that we will not bear the entire cost of remediation of any site to the exclusion of other PRPs who may be jointly and severally liable. The ability of other PRPs to participate has been taken into account, based on our understanding of the parties' financial condition and probable contributions on a per site basis. Additional lawsuits and claims involving environmental matters are likely to arise from time to time in the future.

We incurred $3.2 million, $23.3 million, and $4.4 million of expenses during the years ended December 31, 2017, 2016, and 2015, respectively, for environmental remediation at sites presently or formerly owned or leased by us. As of December 31, 2017 and 2016, we have recorded reserves for environmental matters of $28.9 million and $30.6 million. Of these amounts $8.9 million and $9.6 million, respectively, relate to remediation of sites previously disposed by us. Given the evolving nature of environmental laws, regulations and technology, the ultimate cost of future compliance is uncertain.